

Module 1: Cybersecurity Overview



Topic 3: Data Encryption



What is Data Encryption?

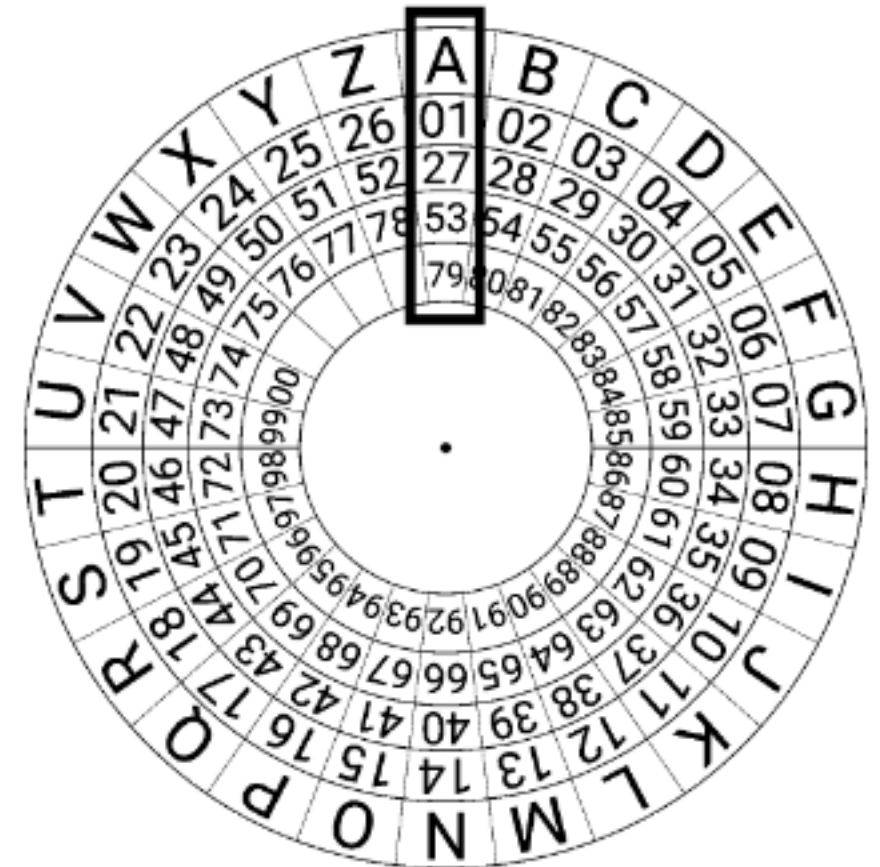
Cryptography refers almost exclusively to encryption, the process of converting ordinary information (plaintext) into ciphertext. Decryption is the reverse, moving from unintelligible ciphertext to plaintext.

plaintext = secret
encrypted = esrcte

The diagram illustrates a simple encryption process. The word 'secret' is shown as the plaintext. Below it, the word 'esrcte' is shown as the encrypted version. Three curved arrows indicate the letter shifts: the 's' in 'secret' moves to the position of the 'e' in 'esrcte', the 'e' moves to the position of the 's', and the 't' moves to the position of the 'c'.

A Cipher (Cypher)

It is a pair of algorithms which creates the encryption and the reversing decryption. The detailed operation of a cipher is controlled both by the algorithm and, in each instance, by a key. This is a secret parameter (ideally, known only to the communicants) for a specific message exchange context.



Why do we need to Encrypt our Data?

- Confidentiality
- Integrity
- Authentication/Authorization

Types of Data Encryption



Ancient Ciphers

The classical algorithms are those invented pre-computer up until around the 1950's.

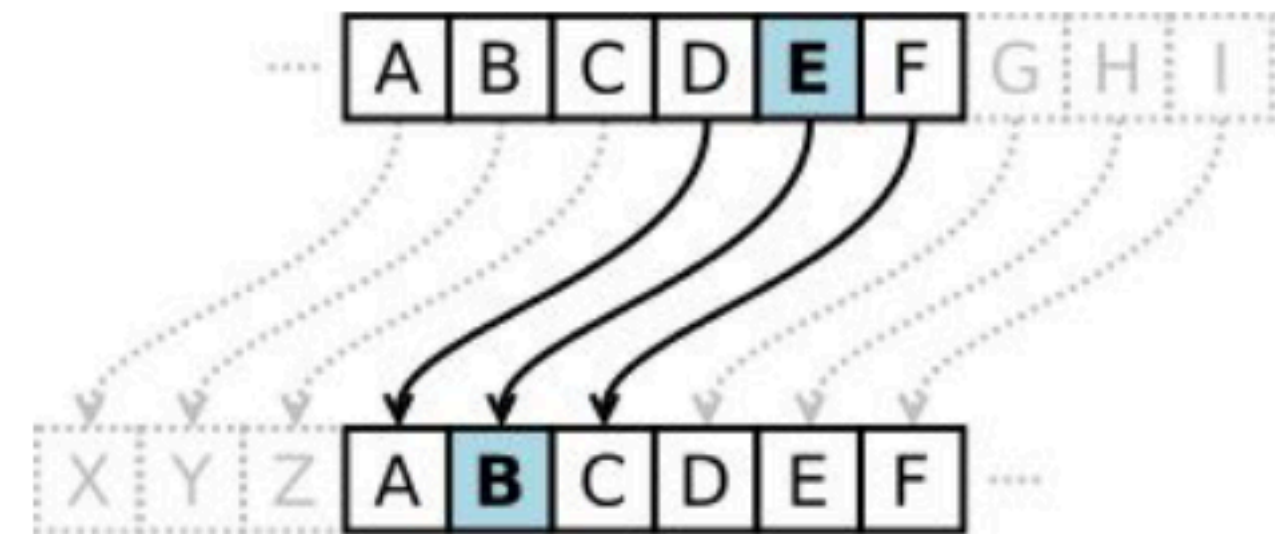


Caesar Ciphers

To pass an encrypted message from one person to another, it is first necessary that both parties have the 'key' for the cipher, so that the sender may encrypt it and the receiver may decrypt it. For the caesar cipher, the key is the number of characters to shift the cipher alphabet.

plaintext: defend the east wall of the castle

ciphertext: efgfoe uif fbtu xbmm pg uif dbtumf



ROT13 Ciphers

The ROT13 cipher is essentially a substitution cipher with a fixed key, if you know the cipher is ROT13, then no additional information is needed to decrypt the message. The substitution key is:

ABCDEFGHIJKLMNOPQRSTUVWXYZ

NOPQRSTUVWXYZABCDEFGHIJKLM

<https://www.dcode.fr/caesar-cipher>

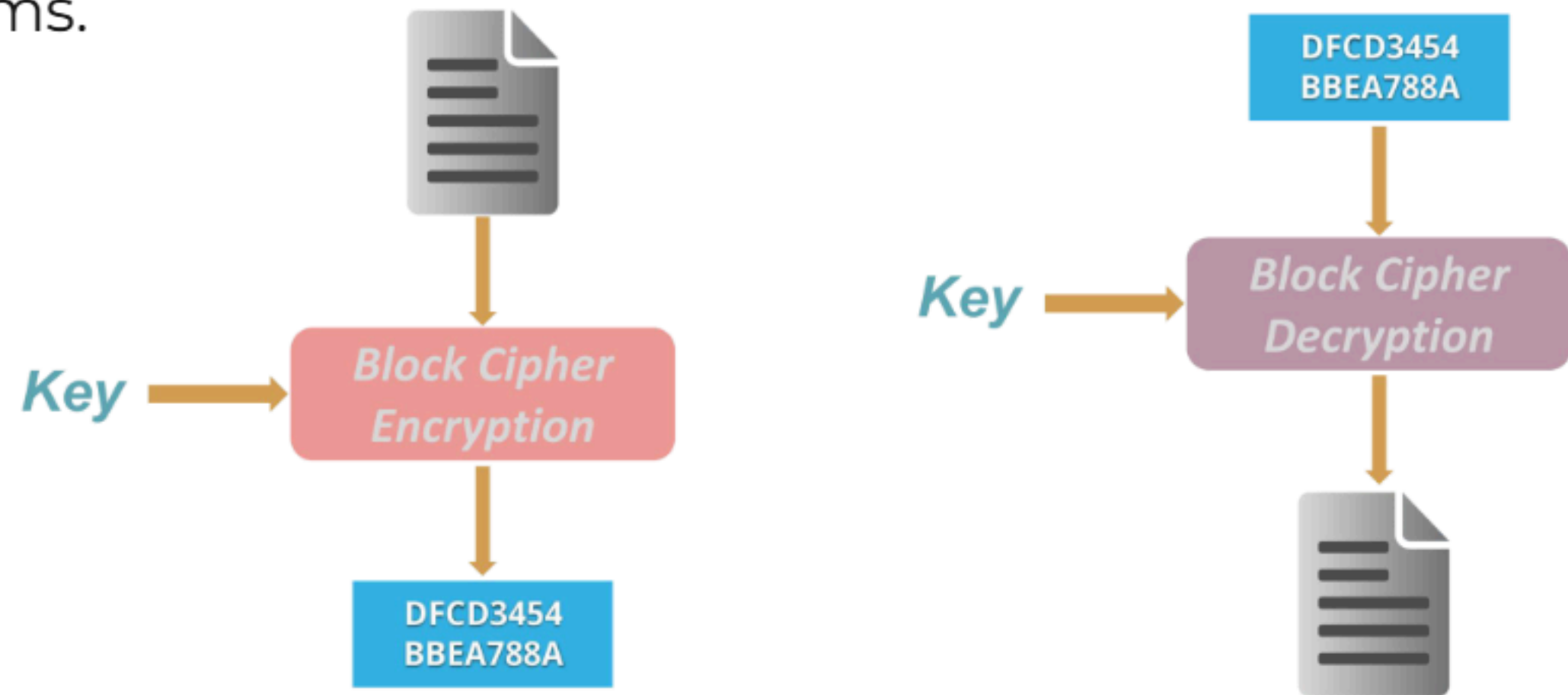
Mechanical Ciphers

Mechanical Ciphers are those that were developed around the second World War, which rely on sophisticated gearing mechanisms to encipher text.



Modern Ciphers

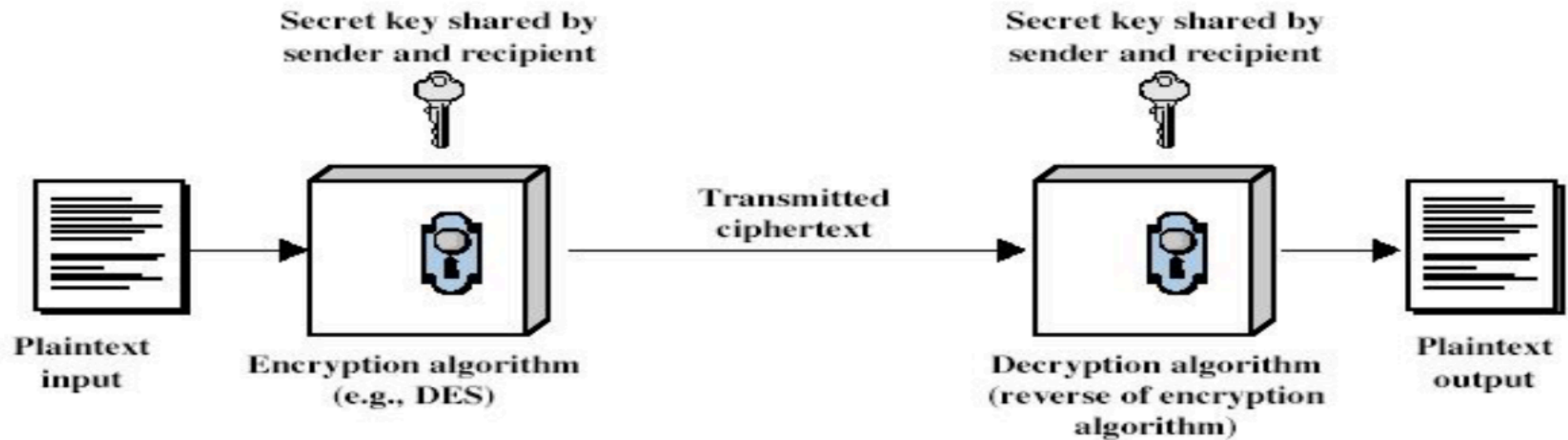
Modern algorithms are those that are used in current technology e.g. block ciphers, public key cryptosystems etc. These algorithms are very secure (otherwise they would not be used), but in many cases we can practice on weakened versions of the algorithms.



Symmetric Vs Asymmetric

- In Symmetric-key encryption the message is encrypted by using a key and the same key is used to decrypt the message which makes it easy to use but less secure.
E.g. AES, DES
- Asymmetric Key Encryption is based on public and private key encryption technique. It uses two different key to encrypt and decrypt the message. It is more secure than symmetric key encryption technique but is much slower.
E.g Diffie-Hellman, ECC, El Gamal, DSA and RSA

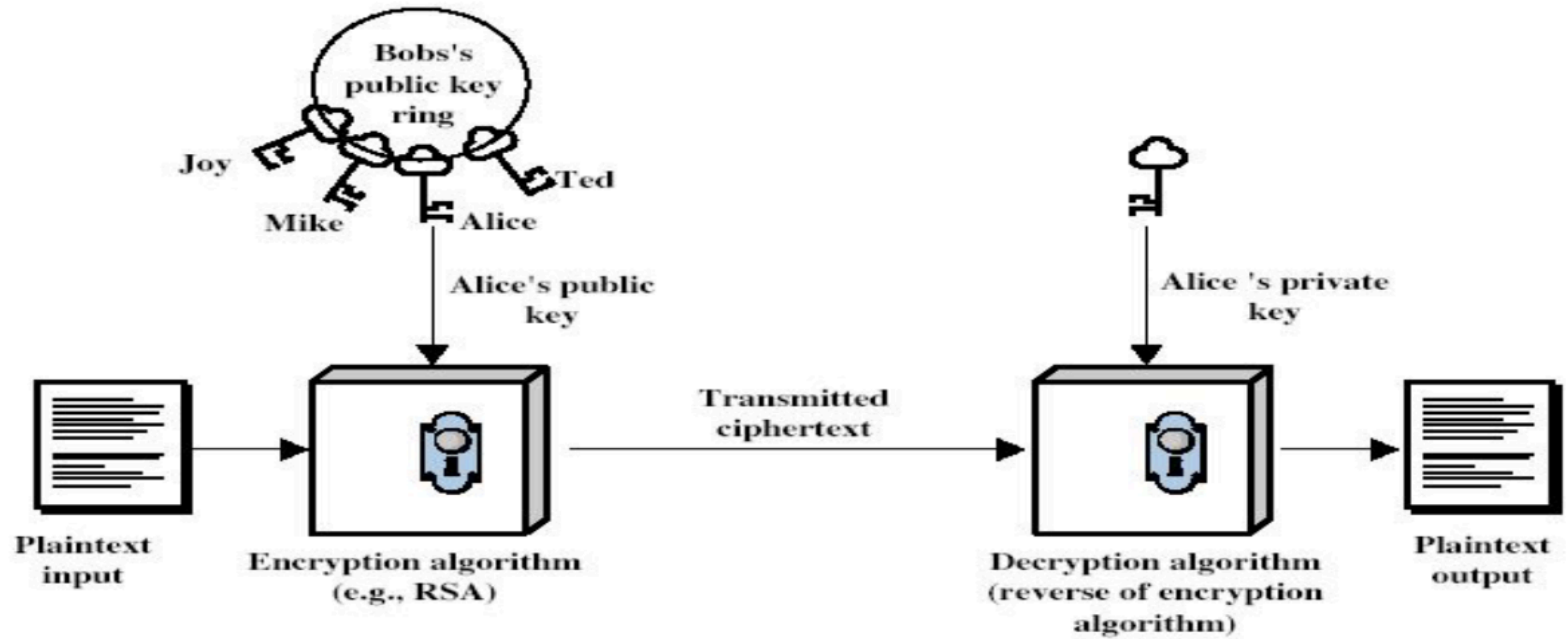
Symmetric Cipher Model



Public Key Cryptography

- **Public-key/two-key/asymmetric** cryptography involves the use of **two** keys:
 - a **public-key**, which may be known by anybody, and can be used to **encrypt messages**, and **verify signatures**
 - a **private-key**, known only to the recipient, used to **decrypt messages**, and **sign** (create) **signatures**
- **Asymmetric** because
 - those who encrypt messages or verify signatures **cannot** decrypt messages or create signatures

Asymmetric Cipher Model



Lab

Hide Data

